Progress Report: Case Study in Statistical Learning

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ABSTRACT

This report applies statistical learning methods to studying the impact of combinations of four kinds of anticancer drugs (Vincristine, Mitoxantrone, Etoposide and Daunorubicin) against leukemia. Various kinds of models are adopted to do feature selection, including Quadratic regression, Ridge regression, Lasso regression, Multi-Layer Perceptron neural network (MLP), as well as Random Forest. The performances of models would be evaluated and compared with each other. We would also give some insights into the problem by discussing the effectiveness as well as drawbacks of the models, and propose some possible improvements.

I. Introduction

i) Background Information

In recent years, the rapid development of machine learning has engaged the interest of people from various fields since machine learning has a wide range of applications due to its strong ability of pattern learning. For example, we could find out how the combination of different medicines would influence the cancer cell viability by statistical learning. That could be quite a difficult problem for classical methods since there are quite a lot features and the relationship between features and outcome may be complicated. Therefore, classical models may result in high variance or bias.

ii) Motivation and Research Plan

Statistical learning, as a popular topic these days, is not only interesting but also highly practical and can be applied to solve various problems. Moreover, compared to other topics, statistical learning is much more interdisciplinary involving knowledge of statistics, mathematical modeling and computer science. All these characteristics are so attractive that I would like to work on it. During this research, I plan to study some general methods to process data as well as models in statistical learning, and apply them to studying cases and solving problems.

II. Methodology

i) Data Processing

The original dataset consists of the dosage levels of Vincristine, Mitoxantrone, Etoposide and Daunorubicin, which are medicines for treating leukemia, as well as the corresponding cancer cell viability after applying the medicines. Since each medicine has four dosage levels, there are totally 256 groups of data. To take the effect of the combination of two drugs into consideration, we select two of the variables with replacement and multiply them together thus create 10 new variables. The new dataset applied for training model consists of 14 features, which are denoted as V, M, E, D, VV, VM, MM, VE, ME, EE, VD, MD, ED, DD, and one outcome, which is cell viability. We also normalize our data by feature scaling. Moreover, to examine the effectiveness of the models, 32 groups of data are randomly selected to serve as test data, else being training data.

ii) Ridge Regression

Instead of least square, ridge regression estimates coefficients $\hat{\beta}^R$ minimizing

$$\sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij})^2 + \lambda \sum_{j=1}^{p}\beta_j^2,$$

where $\lambda \geq 0$ is a tuning parameter. Therefore, ridge regression tries to make residue sum of squares small, while having the effect of shrinking β_j towards zero. Compared to least squares estimate, ridge regression trades off a small increase in bias for a large decrease in variance (James, 2013, p215). It is obvious that different λ would give different estimations of β_j, so we apply 8-folds cross-validation to training data to determine the value of λ minimizing mean squared error (MSE), and then take the value of β_j at this λ to evaluate the effect of 14 features.

iii) Lasso Regression

Lasso regression is similar to ridge regression while using ℓ_1 norm instead of ℓ_2 norm, estimating coefficients $\hat{\beta}^L$ minimizing

$$\sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij})^2 + \lambda \sum_{j=1}^{p}|\beta_j|,$$

where $\lambda \geq 0$ is a tuning parameter. While ridge regression is only able to shrink coefficients towards zero, but not equal to zero, ℓ_1 penalty would force more coefficients to be zero as λ increases. So lasso regression could perform variable selection, and the model we get may contain only a subset of features (James, 2013, p219).

iv) Multi-Linear Perceptron

As shown in Fig.1, a basic unit of a neural net called a perceptron consists of n-vector input, weighted summation, activation function, as well as output. And as shown in Fig.2, MLP consists of an input layer, one or more hidden layers, as well as an output layer. In this paper, we only study MLP with one hidden layer and sigmoid function is set to be activation function in every situation. While training MLP, back propagation algorithm is used to adjust weights to minimize the error $E = \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{d}\|_2$, where $\boldsymbol{y}$ is the output vector we get by calculation, and $\boldsymbol{d}$ is the target output vector. As a model-free estimator, MLP may be able to perform well while the relationship of features and output are sophisticated.

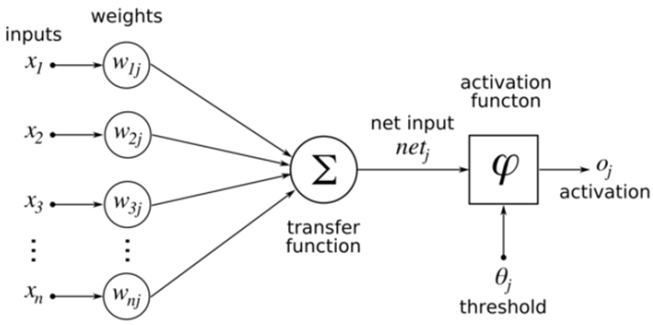

Figure 1: Structure of a simple perceptron (Artificial neural networks/activation functions - Wikibooks, open books for an open world, 2015).

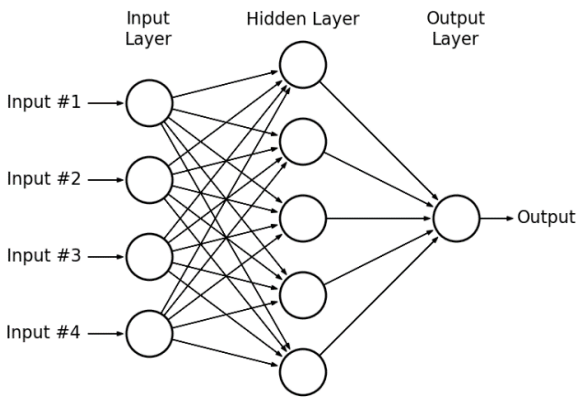

Figure 2: Structure of MLP with single hidden layer (Minnaar, 2015).

v) Random Forest

While training our random forest model, 100 decision trees are constructed, each only taking into consideration a random sample of m features from the full set of features. And while making prediction, random forest takes the mean of predictions of each tree as output. Similar to MLP, if there is a highly non-linear and complex relationship between the features and the response, decision trees may outperform classical approaches. And random forests correct decision trees' habit of overfitting to their training set (Friedman, Hastie, and Tibshirani, 2009, p587-588).

III. Results

Different methods give different results on evaluating the impact of features on decreasing cancer cell viability. Quadratic approximation suggests that V performs best, followed by M, E, VM, D. Fig.3 suggests that while λ is chosen such that MSE is minimized, ridge regression suggests that V performs best, followed by VM, E, VV, EE, and V also outperforms others as for lasso regression, followed by M, E, VM, D. It is also worth noticing that both ridge and lasso regression suggest that in the domain of dosage, cell viability is even an increasing function of M when M is big enough. Moreover, as λ decreases, the order that features enter lasso model is V, VD, E, VM, D, DD, EE, MD, MM, VE, ME, M, VV, ED.

```
> ridge.coef
 (Intercept)            V            M            E            D           VV
 0.894796259 -0.118904130 -0.022453517 -0.063985423 -0.038184543 -0.047688197
          VM           MM           VE           ME           EE           VD
-0.093608825  0.028386012  0.001785041  0.009760485 -0.045140880 -0.033173053
          MD           ED           DD
-0.025496455 -0.013613778 -0.033360242
> lasso.coef
  (Intercept)             V             M             E             D
 0.9403414994 -0.2972474493 -0.1524598017 -0.1248777109 -0.0594308222
           VV            VM            MM            VE            ME
 0.0919237851 -0.0874457688  0.1397286684  0.0330549898  0.0255761968
           EE            VD            MD            ED            DD
-0.0123819454 -0.0121481697 -0.0199385700  0.0005461514 -0.0292457390
```

Figure 3: Coefficients of Ridge and Lasso Model for λ minimizing MSE.

Fig.4 suggests the importance of inputs of our trained MLP estimated by Garson's Algorithm. Fig.5 shows the importance of features in random forest, more important the feature is, higher the value would be.

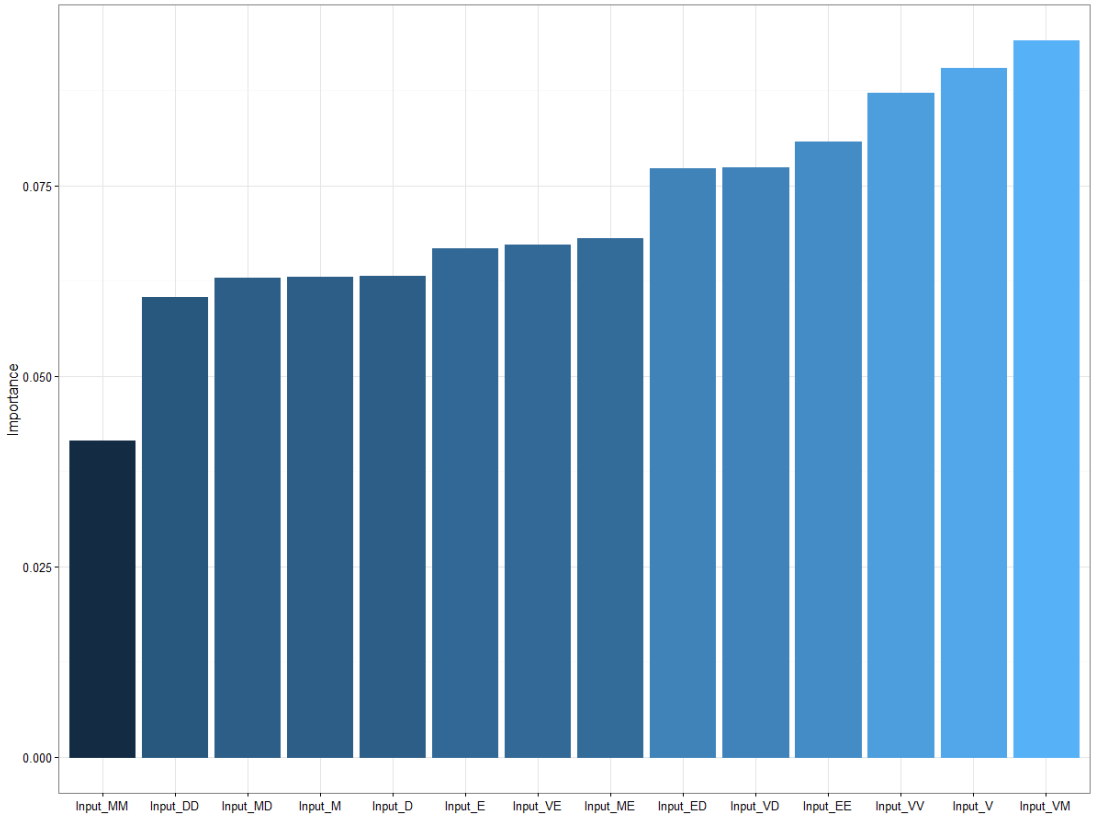

Figure 4: The Importance of Inputs in MLP.

	%IncMSE	IncNodePurity
V	8.10	0.53
M	6.51	0.08
E	6.47	0.13
D	4.92	0.08
VV	10.11	0.62
VM	5.54	0.25
MM	5.37	0.08
VE	11.26	0.42
ME	7.57	0.11
EE	7.26	0.14
VD	8.34	0.37
MD	2.58	0.07
ED	4.73	0.18
DD	6.05	0.09

Figure 5: The Importance of Features in Random Forest.

We measure the performance of different models by calculating the MSE of test data. As Fig.6 shows, quadratic regression performs worst, while lasso regression and random forest outperform other models.

```
> qr.MSE
[1] 0.02724675
> ridge.MSE
[1] 0.004061788
> lasso.MSE
[1] 0.003252355
> mlp.MSE
[1] 0.004595737
> randfore.MSE
[1] 0.003298203
```

Figure 6: MSE of test data of models.

IV. Conclusion

As our results show, random forest and lasso regression performs quite well at capturing the pattern of the impact of combination of medicines on cell viability. The combination of V with other three medicines may be candidates for more effective therapy against leukemia. And the excessive use of M may have negative effect on treating leukemia. Our remaining problem is that the important features suggested by different model varies from one another. As far as I am concerned, improvements could be made in the way of normalization. Models are also to be improved, for

instance, we could combine lasso and ridge regression together, using both ℓ_1 and ℓ_2 penalty at the same time.

REFERENCES

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